smMIP Pipeline Overview

smMIP Analysis Pipeline Overview

Quick Start

```bash
# Basic pipeline (Stages 1-6: FASTQ to mutation calls)
bash /path/to/smMIP-Pipeline/Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/experiment/RAW_FASTQ

# Full pipeline with CHIP analysis (Stages 1-8)
bash /path/to/smMIP-Pipeline/Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/experiment/RAW_FASTQ \
  --chip_analysis
```

Pipeline Stages

Stage	Script	Description
1	BWA.sh	Align paired-end reads to hg19 reference
2	filter_bam.sh	Filter for quality (MAPQ>=50, properly paired)
3	Read_Processing.sh	Map reads to smMIP probes, extract UMIs
4	Level_Bas_Calls.sh	Generate variant pileups at smMIP level
5	Annotate_SNVs.R	Annotate panel with gene/protein/COSMIC info
6	calling_mutations.R	Call mutations using statistical error modeling
7	CHIP_analysis.R	Population-level CHIP analysis (oncoplots, VAF heatmaps)
8	generate_patient_reports.R	Generate individual patient HTML/PDF reports

Command-Line Parameters

Required Parameters

Parameter	Description
`--directory`	Base experiment directory (outputs go here)
`--fastq_dir`	Directory containing raw FASTQ files

Optional Parameters

Parameter	Default	Description
`--panel`	`Universal/Myeloid_Panel_Targets.chr.txt`	Panel design file
`--ref`	`Universal/hg19/hg19.fa`	Reference genome
`--annotated_panel`	`Universal/annotated_Myeloid_Panel_Targets.txt`	Pre-annotated panel
`--config`	`{directory}/config.txt`	Sample configuration file
`--threads`	6	Threads per sample
`--parallel_jobs`	4	Samples to run in parallel
`--use_parallel`	false	Use parallel versions of scripts (flag)
`--force`	false	Force re-run all stages even if outputs exist
`--start`	1	Start at this stage (1-8)
`--stop`	8	Stop after this stage (1-8)

CHIP Analysis Parameters (Stages 7-8)

Parameter	Default	Description
`--chip_analysis`	-	Shorthand for –stop 8 (CHIP analysis stages)
`--chip_vaf`	0.01	

Parameter	Default	Description
		VAF threshold for CHIP filtering
`--chip_pval`	0.05	P-value threshold for CHIP filtering
`--skip_reports`	false	Skip patient report generation (Stage 8)
`--skip_clinvar`	false	Skip ClinVar API queries (faster reports)
`--report_format`	html	Report format: html or pdf

Directory Structure

```
{--directory}/
├── bwa_out/                    # Stage 1 output
│   ├── *.bam
│   ├── *.bam.bai
│   ├── *.flagstat.txt
│   └── _logs/
├── filtered_bam/               # Stage 2 output
│   ├── *.filtered.bam
│   ├── *.filtered.bam.bai
│   └── _logs/
├── Read_Processing/            # Stage 3 output
│   ├── {sample}/
│   │   ├── {sample}_clean.bam
│   │   ├── {sample}_filtered_read_counts.txt
│   │   ├── {sample}_raw_coverage_per_smMIP.txt
│   │   └── {sample}_UMI_usage_per_smMIP.txt
│   ├── pileup/                 # Stage 4 output
│   │   ├── {sample}_raw_pileup.txt
│   │   └── {sample}_sscs_pileup.txt
│   └── _logs/
├── results/                    # Stage 6 output
│   └── called_mutations.txt
├── CHIP_analysis/              # Stages 7-8 output
│   ├── CHIP_oncoplot.pdf/png
│   ├── VAF_heatmap.pdf/png
│   ├── VAF_dotplot.pdf/png
│   ├── co_mutation_heatmap.pdf
│   ├── gene_mutation_summary.csv
│   ├── variant_annotation_summary.csv
│   ├── co_mutation_fisher_results.csv
│   └── patient_reports/
│       ├── {sample}_CHIP_Report.html
│       └── annotation_cache/
└── config.txt                  # Sample configuration
```

Pipeline Flow Diagram

```
FASTQ files (*.R1*.fastq.gz, *.R2*.fastq.gz)
    │
    ▼
┌──────────────────────────────────────────────────────────────┐
│ Stage 1: BWA Alignment                                       │
│ Script: BWA.sh / BWA_parallel.sh                             │
│ Input:  *R1*.fastq.gz, *R2*.fastq.gz                         │
│ Output: *.bam, *.bam.bai, *.flagstat.txt                     │
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│ Stage 2: BAM Filtering                                       │
│ Script: filter_bam.sh / filter_bam_parallel.sh               │
│ Input:  *.bam from Stage 1                                   │
│ Output: *.filtered.bam                                       │
│ Filter: MAPQ >= 50, properly paired (-f 2)                   │
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┌──────────────────────────────────────────────────────────────┐
│ Stage 3: Read Processing / UMI Extraction                    │
│ Script: Read_Processing.sh / Read_Processing_parallel.sh     │
│ R Script: map_smMIPs_extract_UMIs.R                          │
│ Input:  *.filtered.bam + Panel_Targets.txt                   │
│ Output: Per sample directory containing:                     │
│         - {sample}_clean.bam                                 │
│         - {sample}_filtered_read_counts.txt                  │
│         - {sample}_raw_coverage_per_smMIP.txt                │
│         - {sample}_UMI_usage_per_smMIP.txt                   │
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│ Stage 4: Pileup Generation (Level Base Calls)                │
│ Script: Level_Bas_Calls.sh / Level_Bas_Calls_parallel.sh     │
│ R Script: smMIP_level_raw_and_consensus_pileups.R            │
│ Input:  {sample}_clean.bam + Panel_Targets.txt               │
│ Output: - {sample}_raw_pileup.txt                            │
│         - {sample}_sscs_pileup.txt (consensus)               │
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│ Stage 5: Panel Annotation (ONE-TIME per panel)               │
│ R Script: Annotate_SNVs.R                                    │
│ Input:  Panel_Targets.txt                                    │
│ Output: annotated_Panel_Targets.txt                          │
│ Annotations: gene, protein, COSMIC, MAF, CADD scores         │
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```

```
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│ Stage 6: Mutation Calling                                    │
│ R Script: calling_mutations.R                                │
│ Input:  - Pileup folder (*_raw_pileup.txt, *_sscs_pileup)    │
│         - Configuration file (sample info)                   │
│         - Annotated panel file                               │
│ Output: called_mutations.txt                                 │
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    │
    ▼
  called_mutations.txt
    │
    ▼ (Optional: --chip_analysis)
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│ Stage 7: CHIP Population Analysis                            │
│ R Script: CHIP_analysis.R                                    │
│ Input:  called_mutations.txt                                 │
│ Output: - CHIP_oncoplot.pdf/png (mutation landscape)         │
│         - VAF_heatmap.pdf/png (allele frequency matrix)      │
│         - VAF_dotplot.pdf/png (VAF distribution)             │
│         - co_mutation_heatmap.pdf (gene co-occurrence)       │
│         - gene_mutation_summary.csv                          │
│         - variant_annotation_summary.csv                     │
│         - co_mutation_fisher_results.csv                     │
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│ Stage 8: Patient Report Generation                           │
│ R Script: generate_patient_reports.R                         │
│ Input:  called_mutations.txt                                 │
│ Output: - patient_reports/{sample}_CHIP_Report.html          │
│         Per-patient clinical-style reports with:             │
│         - CHIP gene mutation summary                         │
│         - ClinVar annotations                                │
│         - Clinical significance assessment                   │
│         - VAF visualizations                                 │
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    ▼
  Patient Reports (HTML/PDF)
```

Configuration File Format

The configuration file (`config.txt`) describes sample metadata for mutation calling.

Format

Tab-delimited file with columns:

Column	Description
id	Sample ID (must match pileup filenames)
type	Sample type: `case` or `control`
replicate	Technical replicate pairing (or `NA`)

Example

```
id  type    replicate
Sample1 case    NA
Sample2 case    NA
Sample3 control NA
Sample4 case    rep1
Sample5 case    rep1
```

Auto-Generation

If `--config` is not provided, the pipeline will: 1. Scan the pileup directory for `*_raw_pileup.txt` files 2. Extract sample IDs from filenames 3. Generate a template `config.txt` with all samples as `case` 4. **Stop and prompt user to edit the file** 5. Re-run with `--start 6` to continue

Mutation Calling Parameters

Key parameters for `calling_mutations.R`:

Parameter	Default	Description
`-s`	Required	Path to pileup folder
`-f`	Required	Configuration file (sample layout)
`-a`	Required	Annotated panel file
`-b`	"sum"	Binomial error rate method ("sum" or "max")
`-g`	10	Min coverage for overlap (R1/R2)
`-m`	0.001	MAF cutoff to exclude known SNPs
`-v`	0.05	VAF cutoff for error modeling
`-p`	0.05	P-value cutoff (Bonferroni corrected)
`-d`	2	Fold-difference for cross-sample VAF
`-t`	1	Number of threads

Output: called_mutations.txt

Key columns in the final mutation calls:

Column	Description
sample_ID	Sample identifier

Column	Description
smMIP	smMIP probe name
chr	Chromosome
pos	Position (hg19)
ref	Reference allele
alt	Alternative allele
gene	Gene symbol
protein	Protein change (e.g., p.V600E)
cosmic	COSMIC ID if known
maf	Minor allele frequency (population)
variant_type	SNV, insertion, deletion
cadd_scaled	CADD pathogenicity score
P-value	Raw p-value
P-value.Bonferroni	Bonferroni-corrected p-value
non.ref.counts	Non-reference allele counts
total.depth	Total read depth
allele.frequency	Variant allele frequency
SSCS.*	Consensus-based metrics
flags	Quality flags

Output: CHIP Analysis (Stages 7-8)

Stage 7 Outputs

File	Description
`CHIP_oncoplot.pdf/png`	Mutation landscape visualization showing genes x samples
`VAF_heatmap.pdf/png`	Heatmap of variant allele frequencies
`VAF_dotplot.pdf/png`	Distribution of VAFs per gene
`co_mutation_heatmap.pdf`	Gene co-occurrence/mutual exclusivity analysis
`gene_mutation_summary.csv`	Frequency statistics per gene
`variant_annotation_summary.csv`	Per-variant details with protein changes
`co_mutation_fisher_results.csv`	Fisher's exact test results for co-mutation

Stage 8 Outputs

File	Description
`patient_reports/{sample}_CHIP_Report.html`	Individual patient reports with:

File	Description
	- Clinically significant mutations (VAF ≥ 1%) - Emerging clones (VAF < 1%) - CHIP gene clinical summaries - ClinVar pathogenicity annotations - VAF visualizations

CHIP Analysis Parameters

Parameter	CLI Flag	Default	Description
VAF threshold	`--chip_vaf`	0.01	Minimum VAF for CHIP filtering
P-value threshold	`--chip_pval`	0.05	Maximum p-value for filtering
Skip ClinVar	`--skip_clinvar`	false	Use local knowledge base only
Report format	`--report_format`	html	Output format: html or pdf

Usage Examples

Full Pipeline Run

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs
```

Parallel Processing (Recommended for Large Datasets)

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --use_parallel \
  --parallel_jobs 4 \
  --threads 6
```

Resume from Specific Stage

```bash
# After editing config.txt, run only mutation calling
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --start 6 --stop 6
```

Force Re-run All Stages

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --force
```

Run CHIP Analysis on Existing Mutations

```bash
# Run only CHIP stages (7-8) on existing called_mutations.txt
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --start 7 --stop 8
```

Full Pipeline with CHIP Analysis

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --chip_analysis \
  --report_format html
```

CHIP Analysis with Custom Thresholds

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --start 7 --stop 8 \
  --chip_vaf 0.02 \
  --chip_pval 0.01 \
  --skip_clinvar  # Faster, uses local knowledge base only
```

Custom Panel and Reference

```bash
bash Master_pipeline.sh \
  --directory /path/to/experiment \
  --fastq_dir /path/to/fastqs \
  --panel /path/to/custom_panel.txt \
  --ref /path/to/hg38.fa \
  --annotated_panel /path/to/annotated_panel.txt
```

Resource Requirements

Memory

Stage	Memory per Sample	Notes
1 (BWA)	~8 GB	Scales with reference genome size
2 (Filter)	~4 GB	I/O bound

Stage	Memory per Sample	Notes
3 (ReadProc)	~30 GB	R memory-intensive
4 (Pileup)	~25 GB	R memory-intensive
5 (Annotate)	~8 GB	Network-bound (API calls)
6 (Mutations)	~16 GB	Depends on cohort size
7 (CHIP)	~16 GB	Scales with cohort size
8 (Reports)	~8 GB	Per-sample, ClinVar API calls

Recommended Settings

System RAM	–parallel_jobs	–threads
32 GB	2	4
64 GB	3	6
128 GB	4	6
256 GB	8	6

Dependencies

Command-Line Tools

- bwa (or bwa-mem2)
- samtools
- GNU parallel (for –use_parallel)

R Packages

Core Pipeline: - optparse - data.table - parallel - cellbaseR (for annotation) - IRanges - Rsamtools

CHIP Analysis (Stages 7-8): - tidyverse - ComplexHeatmap (Bioconductor) - circlize - RColorBrewer - rmarkdown - knitr - kableExtra - rentrez (for ClinVar API)

Troubleshooting

Out of Memory

- Reduce `--parallel_jobs`
- Ensure R_MAX_VSIZE is set (automatically handled by pipeline)

Missing R1/R2 Pairs

- Check FASTQ naming convention matches `*R1*.fastq.gz` pattern
- Use `--r1_glob` and `--r2_token` to customize pattern

Stage Skip Not Working

- Use `--force` to override skip logic
- Check that expected output files exist and are non-empty

Config File Issues

- Ensure tab-delimited format
- Sample IDs must match pileup filenames exactly
- All samples need a type assignment (case or control)